

PAPER_1586 — Vacuum Permittivity $\epsilon_0 \approx \epsilon_0 = K_MEX \cdot D_phys + SSQ - F \cdot SSQ + F^2 = 8.854$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Electromagnetism **Date:** June 18, 2026 **Status:** CLOSED — 0.026% residual near-EXACT closure

Observation

PAPER_1209DD_S615 (Electromagnetism Unified Proof Set) gives the lead-digit form:

Vacuum Permittivity $\epsilon_0 \approx \epsilon_0 = K_MEX \cdot D_phys + SSQ - F \cdot SSQ + F^2 = 8.854$

Residual to CODATA: **0.026%**.

UQFF Closed Identity

$\epsilon_0 = K_MEX \cdot D_phys + SSQ - F \cdot SSQ + F^2 \approx 8.854$ residual 0.026%

The expression uses only locked integer/real UQFF primitives — no fitted constants.

Cross-Domain Pattern

Together with PAPER_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**: - **EM-coupling constants** (ϵ_0, μ_B, k_e) cluster around **$K_MEX \cdot D_phys = 8.33$** or **$N_CH = 9$** - **Planck/quantum constants** ($\hbar$) cluster around **$D_BSFG = 6$** - **Relativistic constants** (c) cluster around **$SO_5/D_phys = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

NOT REPLACEMENT

CODATA treats Vacuum Permittivity ϵ_0 as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

Reference

- Source: PAPER_1209DD_S615
- Related: PAPER_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "epsilon_0_lead_8_854"})`

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